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Project 003



Project Introduction

PitchSense

"What are the real odds, right now?"

Ball-by-Ball Win Probability Engine

PitchSense

The Problem

Why is predicting a cricket winner so difficult?



The game can change in just a few deliveries!

It depends on a constantly changing match state:

- Runs Needed
- Balls Remaining
- Wickets in Hand
- Current Run Rate
- Required Run Rate

So, relying only on instinct or commentary often isn't enough.

PitchSense

The Solution

PitchSense

A data-driven win probability engine for T20 cricket.



What is PitchSense?

An ML-powered platform that predicts the chasing team's **probability of winning** after every delivery in a T20 match.

How does it solve the problem?

IPL Data → Feature Engineering → Logistic Regression → Win Probability

Why is ball-by-ball probability interesting?

- Captures real-time momentum shifts
- Reveals the impact of key events (4s, 6s, wickets)
- Shows how the game can swing in just a few deliveries

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PitchSense

How Does It Work?

From historical IPL data to real-time win probability — a simple and explainable **ML pipeline.**

Cricsheet IPL Data
Historical ball-by-ball match data (JSON)

→

Data Processing
Clean, flatten & structure match data

→

Feature Engineering
Create match-state features (runs, balls, wickets, run rates)

→

Logistic Regression
Learn the relationship between match state and win outcome

→

Win Probability Prediction
P(Chasing Team Wins) for every delivery

Key Match-State Features

- Runs Needed
- Balls Remaining
- Wickets in Hand
- Current Run Rate (CRR)
- Required Run Rate (RRR)
- Innings Progress

The Model: Logistic Regression

$$P(win \mid X) = \frac{1}{1 + e^{-(w_0 + \sum w_i x_i)}}$$

- Classification model (Win / Loss)
- Outputs a calibrated win probability (0 – 1)
- Trained on 1,195 IPL matches (138,683 deliveries)

Output

P(Chasing Team Wins)

65.7%

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PitchSense

Project Highlights

A quick look at the tech, features, and results of PitchSense.

Technologies Used

- **Next.js** (Frontend)
- **TypeScript**
- **Python** (ML & Data)
- **scikit-learn**
- **FastAPI** (Backend)
- **Tailwind CSS**
- **Vercel** (Deployment)
- **GitHub** (Version Control)

Main Capabilities

- Ball-by-ball win probability prediction
- Historical IPL match explorer & replay
- Win probability trajectory visualization
- Interactive scenario simulator (what-if analysis)
- Model evaluation & feature insights

ML Result (Logistic Regression)

77.97%
Accuracy

0.4461	0.8734	0.1469
Log-Loss	ROC AUC	Brier Score

Outperforms the heuristic baseline (0.5437 log-loss, 72.17% accuracy).

Let's Explore PitchSense!